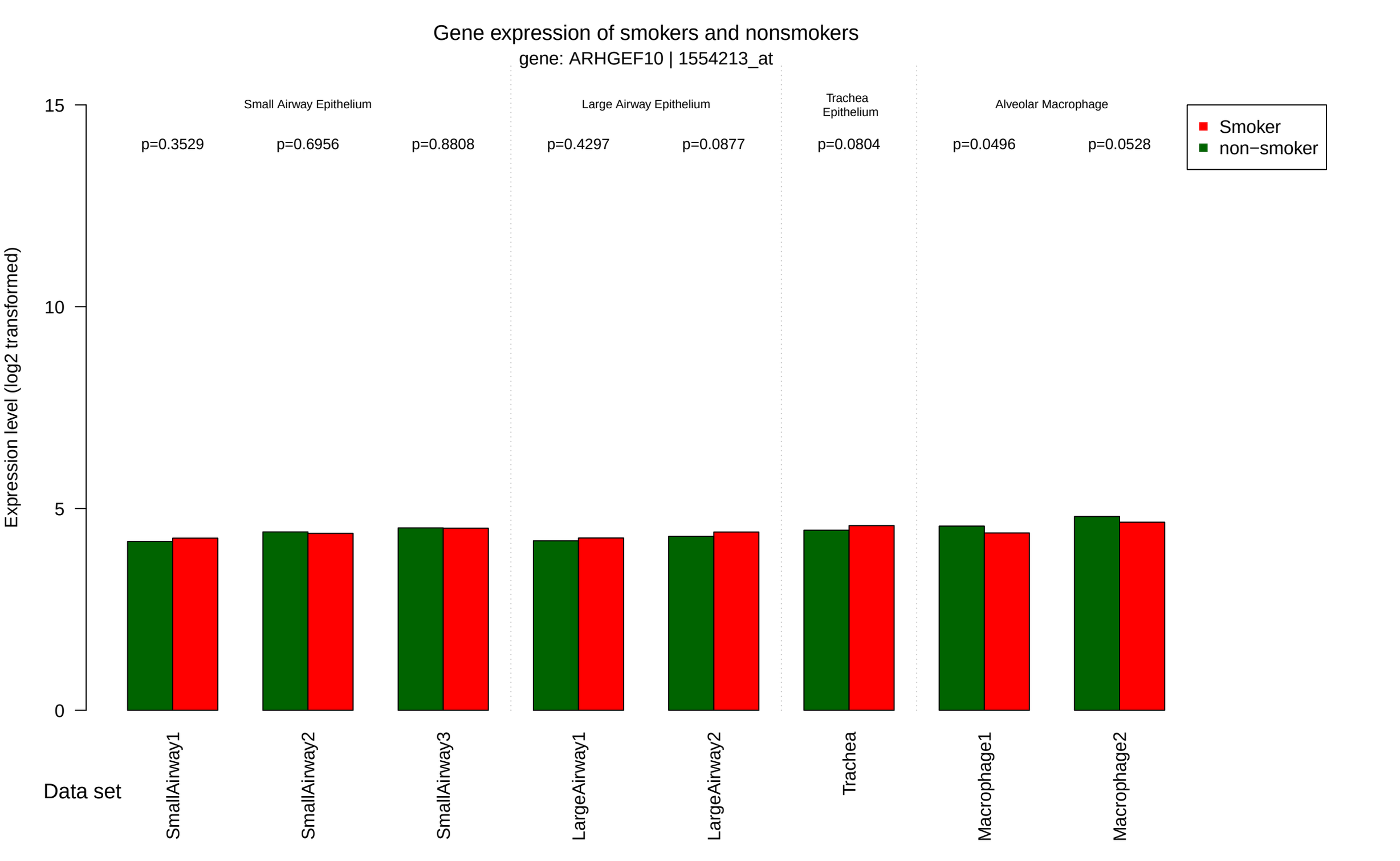
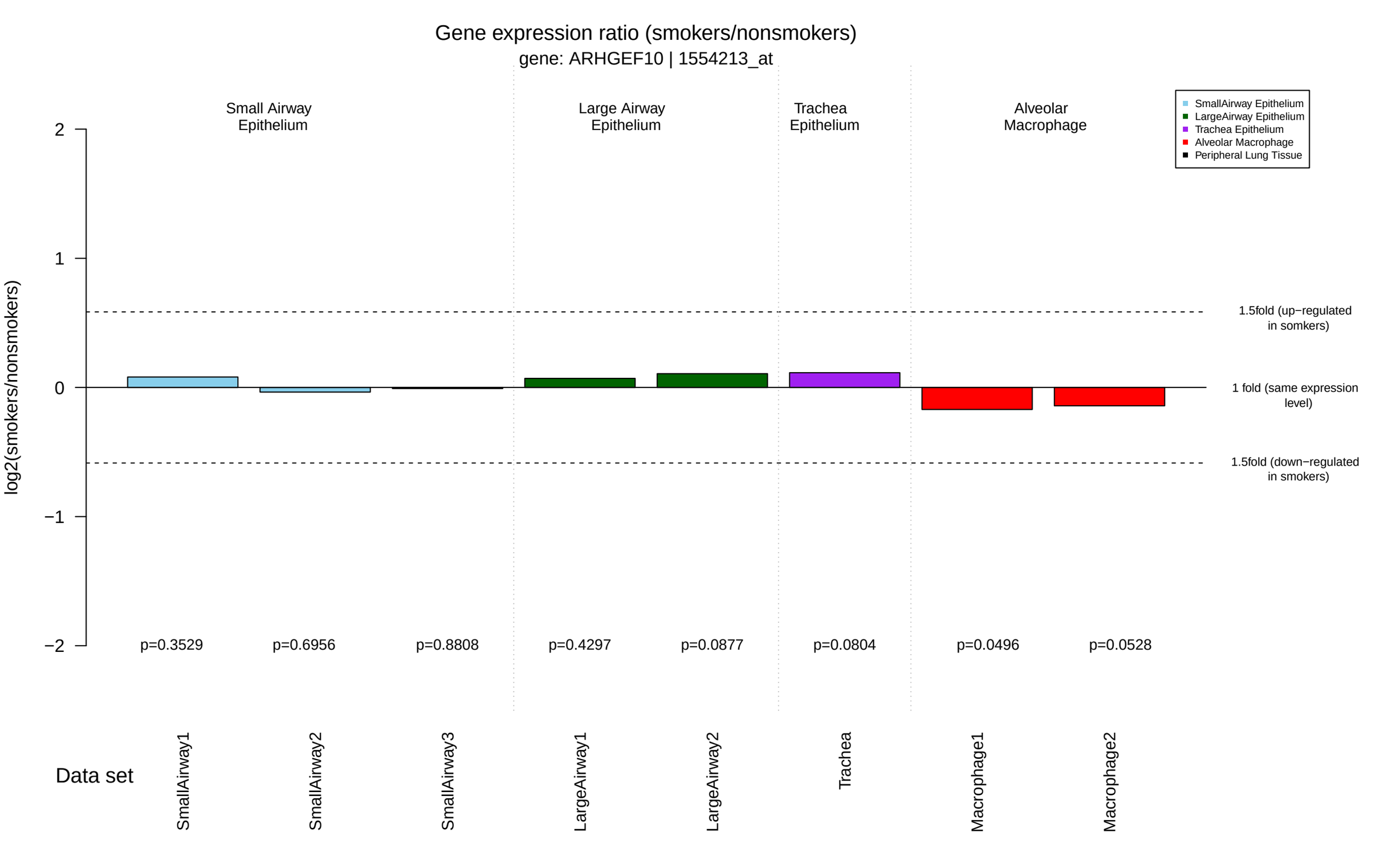
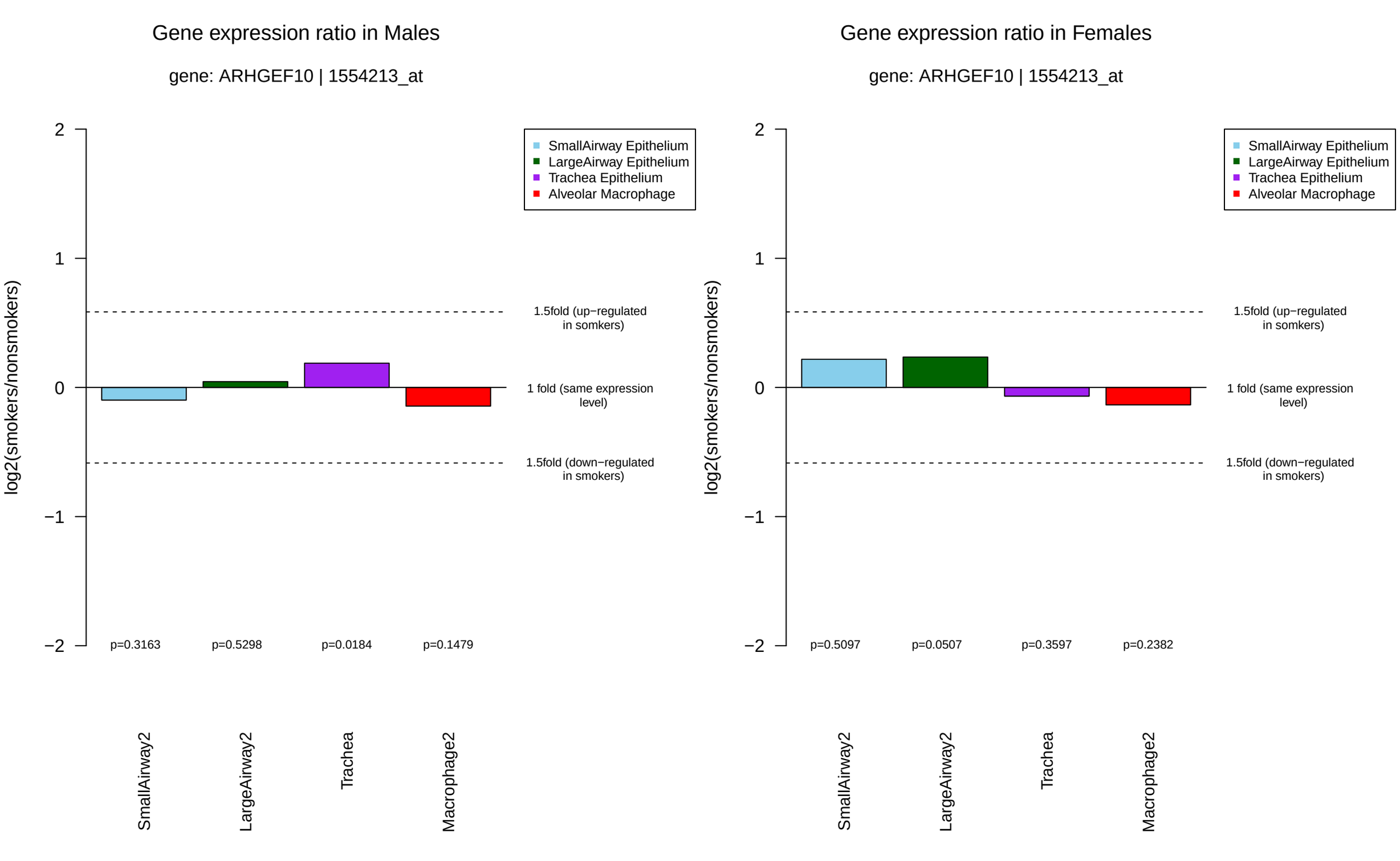
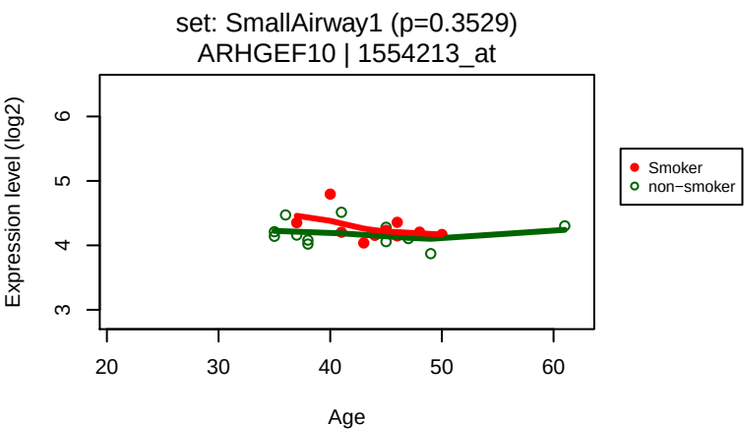
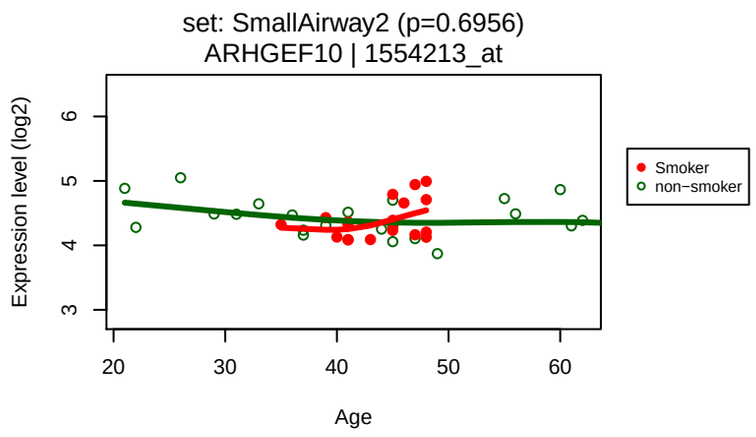
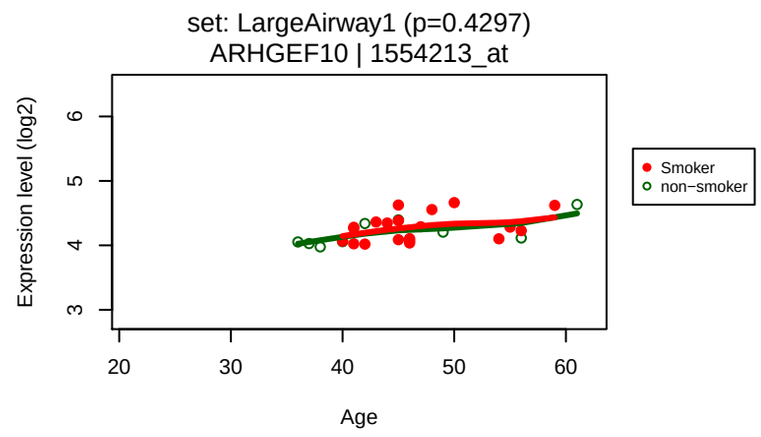

Gene expression of smokers and nonsmokers
gene: ARHGEF10 | 1554213_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3529
p=0.6956
p=0.8808
p=0.4297
p=0.0877
p=0.0804
p=0.0496
p=0.0528
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

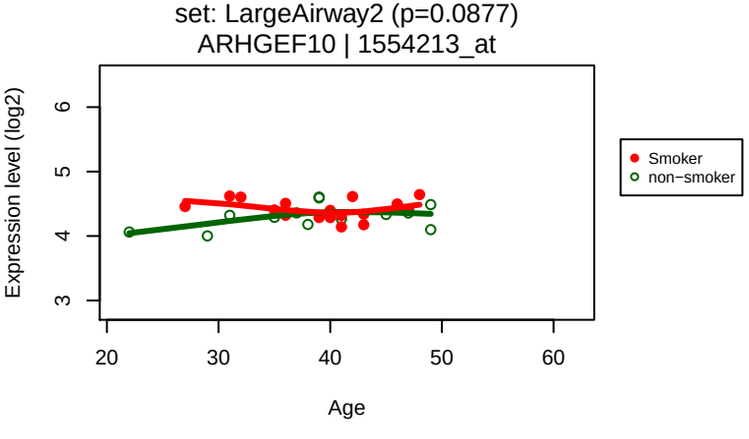

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF10 | 1554213_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3529
p=0.6956
p=0.8808
p=0.4297
p=0.0877
p=0.0804
p=0.0496
p=0.0528
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

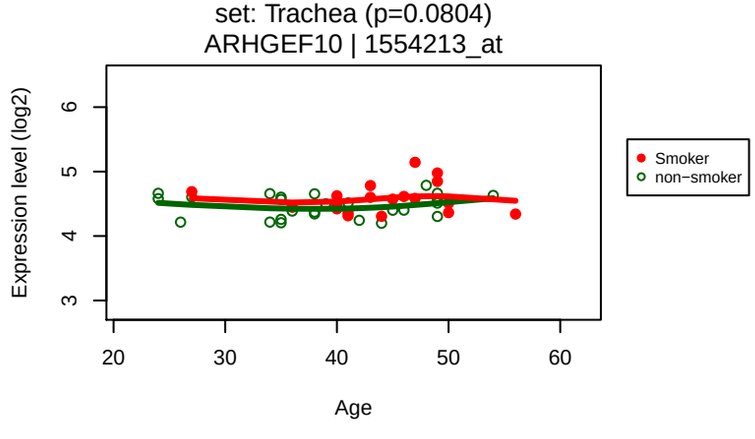

Gene expression ratio in Males
gene: ARHGEF10 | 1554213_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3163
p=0.5298
p=0.0184
p=0.1479
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF10 | 1554213_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5097
p=0.0507
p=0.3597
p=0.2382
SmallAirway2
LargeAirway2
Trachea
Macrophage2

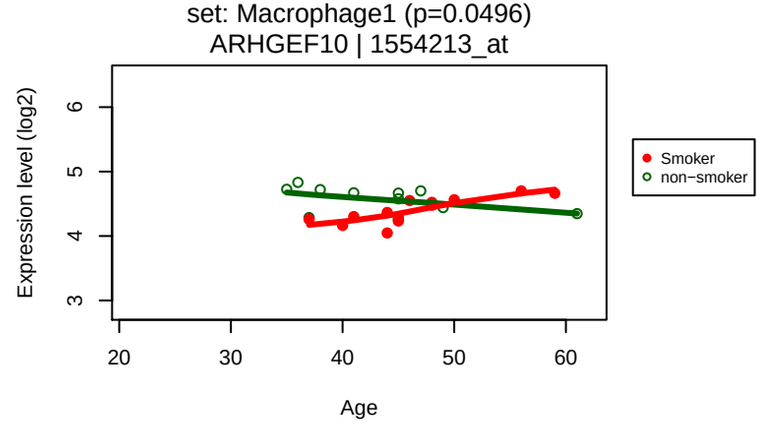

set: SmallAirway1 (p=0.3529)
ARHGEF10 | 1554213_at
Expression level (log2)
Age
Smoker
non−smoker

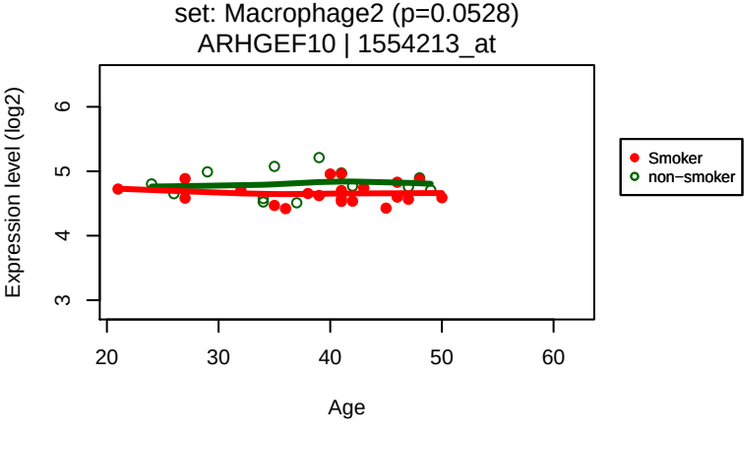

set: SmallAirway2 (p=0.6956)
ARHGEF10 | 1554213_at
Expression level (log2)
Age
Smoker
non−smoker

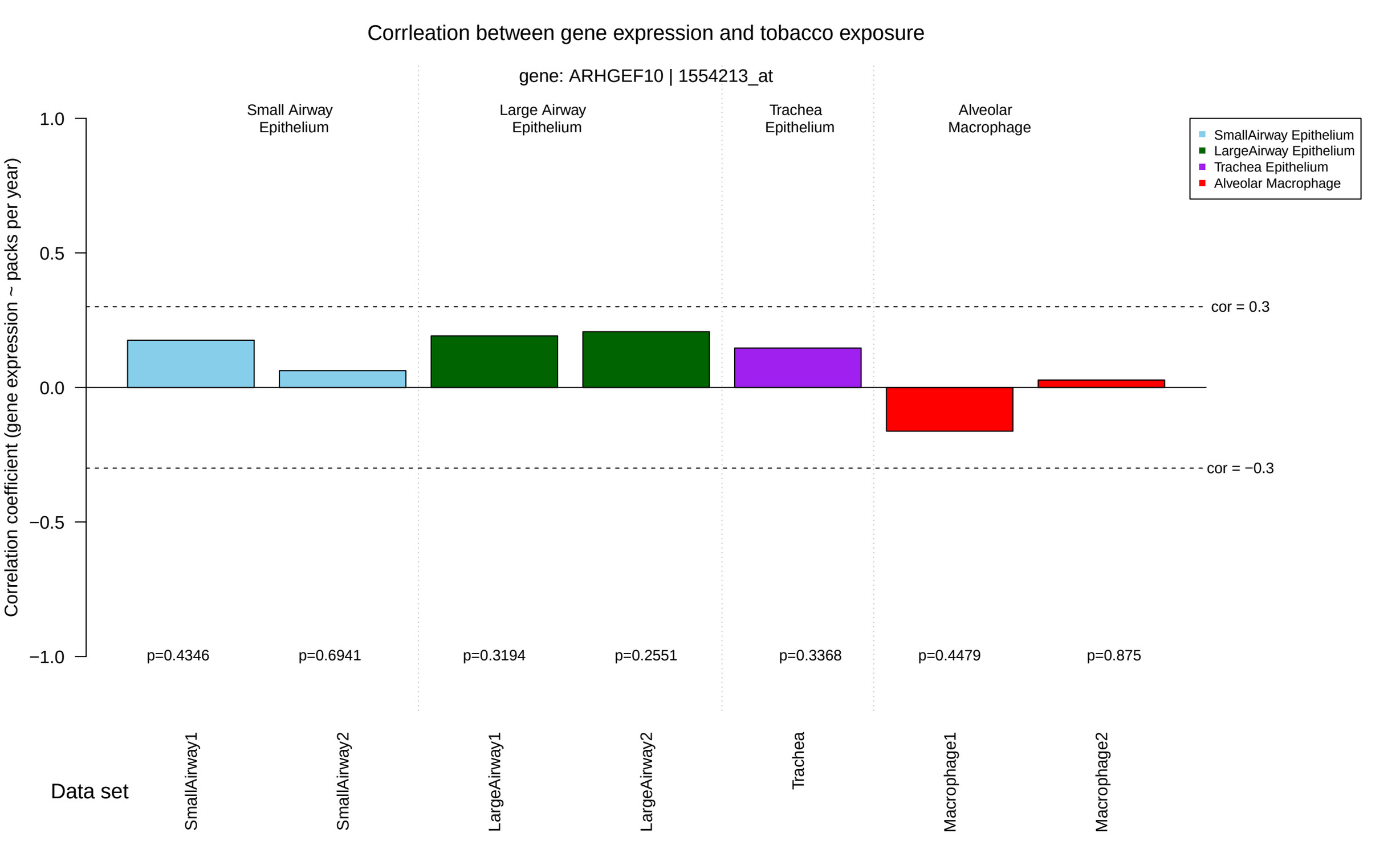

set: LargeAirway1 (p=0.4297)
ARHGEF10 | 1554213_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0877)
ARHGEF10 | 1554213_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0804)
ARHGEF10 | 1554213_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0496)
ARHGEF10 | 1554213_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0528)
ARHGEF10 | 1554213_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF10 | 1554213_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4346
p=0.6941
p=0.3194
p=0.2551
p=0.3368
p=0.4479
p=0.875
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2